

NVIDIA Cosmos, Explained in Depth

A beginner-to-intermediate guide to all four generations — Cosmos 1, 2, 2.5, and 3 — with diagrams, model sizes, data, inference speed, and the pretraining + post-training (SFT / GRPO) recipes.

What is a “World Foundation Model” (WFM)?

A **language** model learns from text and predicts the next *word*. A **World Foundation Model** learns from video and predicts what happens next in the *physical world* — how things move, collide, and behave. NVIDIA **Cosmos** is a family of WFMs for **Physical AI** (robots, self-driving cars): instead of testing a robot millions of times in reality, engineers let Cosmos **imagine** realistic video of what would happen and train on that.

Mental model: Cosmos is a “dream engine” for machines. Each generation dreams with higher quality, runs faster, understands your request better, and — by Cosmos 3 — even hears sound and decides physical actions.

Mini-glossary (so the rest makes sense)

Tokenizer	Compresses video into a small set of numbers (“tokens”) the model can handle, and decompresses them back to pixels.
Diffusion	Starts from random noise and repeatedly “cleans” it into a video. High quality, but many steps = slower.
Autoregressive (GPT-style)	Generates the video one chunk at a time, like typing word by word. Fast, good for real-time.
Flow-based	A cleaner, more direct cousin of diffusion: learns a straight “flow” from noise to data, often with fewer steps.
DiT	“Diffusion Transformer” — the Transformer architecture that powers the diffusion engine.
Mixture-of-Transformers	Several specialized Transformers in one model (e.g. one that <i>reasons</i> + one that <i>generates</i>).
Pretraining	Learning general knowledge from a huge, broad dataset.
SFT	Supervised Fine-Tuning: teach the model on curated question→answer examples after pretraining.
GRPO	An RL method: the model tries an answer several times; better-than-average tries get reinforced. Sharpens reasoning.

The big picture: all four generations side by side

	Cosmos 1	Cosmos 2	Cosmos 2.5	Cosmos 3
Released	Jan 2025	Jun 2025	Sep 2025	Jun 2026
Core engine	TWO engines: diffusion + GPT-style	Refined diffusion	ONE unified flow-based model	Mixture-of-Transformers (reason + generate)
Inputs	Text, or image/video	Text, or image/video	Text/image/video (one model)	Text, image, video, AUDIO + ACTIONS
Outputs	Future video	Video (480/704p)	Video up to 30s, multi-camera	Video + actions + audio, with reasoning
Reads text via	T5-XXL encoder	T5-style encoder	Cosmos-Reason1 (VLM)	Built-in reasoning transformer
Model sizes	Diff 7B/14B, GPT 4B/12B	0.6B / 2B / 14B	2B / 14B (+auto/robot variants)	Super / Nano / Edge
Training data	~100M clips (from 20M hrs)	+ action datasets	200M curated clips	~20 trillion tokens
Post-training	Video2World fine-tune	Action-conditioned fine-tune	Model merge + RL	SFT + RL (GRPO-style)
One-word vibe	Kitchen sink	Polish	Unify	Leap

Table 1. Each column is colour-coded and reused throughout this guide: [Cosmos 1](#), [Cosmos 2](#), [Cosmos 2.5](#), [Cosmos 3](#).

How the architecture changed (diagrams)

Cosmos 1 — two parallel engines

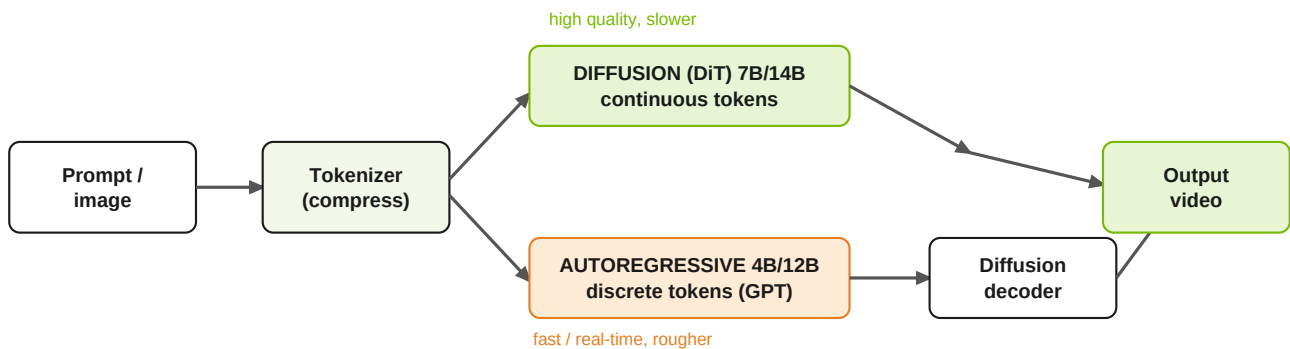


Figure 1. Cosmos 1 shipped two separate engines. Diffusion (top) = best quality; autoregressive (bottom) = fast, but needs a diffusion decoder to clean up its output. Text enters both via a T5-XXL encoder (not shown).

Cosmos 2.5 — one unified flow model with a “smart” text encoder

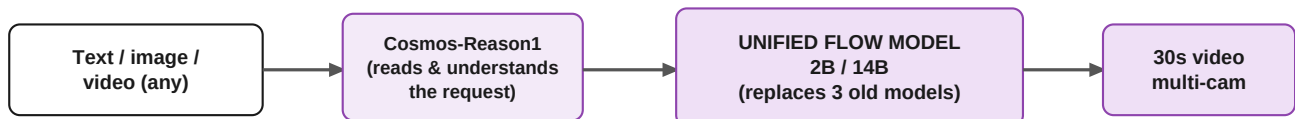


Figure 2. Cosmos 2.5 merged the three separate task-models of earlier versions into a single flow-based model, and upgraded the text encoder to Cosmos-Reason1 — a model that actually understands physics and language.

Cosmos 3 — mixture-of-transformers (thinks before it dreams)

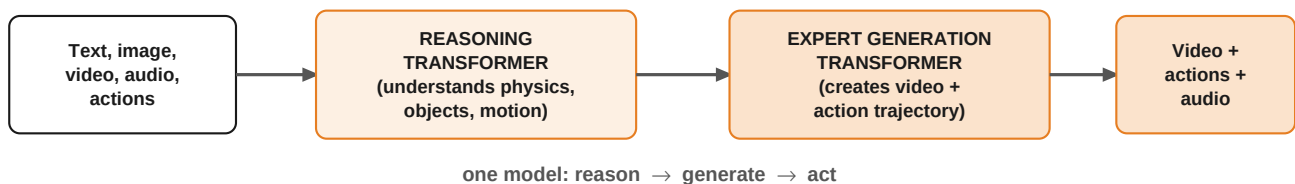


Figure 3. Cosmos 3 is no longer just a video predictor. A reasoning transformer first understands the scene, then an expert generation transformer produces video AND physical actions (and audio) — all in one “omnimodel”.

The data: scale and diversity

A WFM is only as good as the video it learns from. Cosmos 1 curated **~100 million clips** out of **20 million hours** of raw video, deliberately balanced across nine kinds of physical scene so the model isn't biased toward one (e.g. only driving). Later generations grew the data by orders of magnitude.

Cosmos 1 training-data mix (by category)

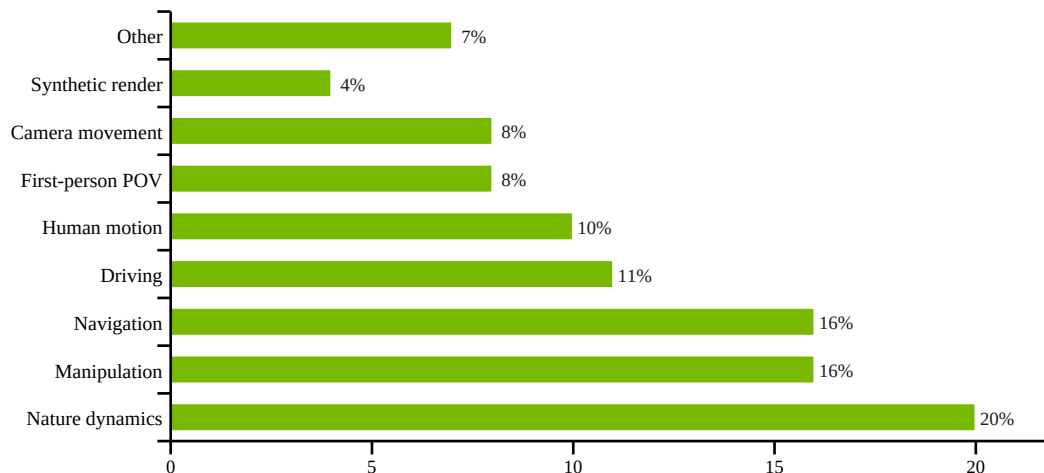


Figure 4. Diversity is intentional: nature, robot manipulation, and navigation dominate, but driving, human motion and synthetic data are all represented. This breadth is what lets one pretrained model be fine-tuned for many downstream robots/vehicles.

Data scale across generations

Generation	Pretraining data	Notes
Cosmos 1	~100M clips (from ~20M hours)	9 balanced categories; ~30% removed by dedup
Cosmos 2	Cosmos 1 data + action datasets	Bridge, AgiBotWorld, GR00T Dreams for post-training
Cosmos 2.5	200M curated high-quality clips	+ domain post-training data (auto, robotics)
Cosmos 3	~20 trillion tokens	~1B images, ~400M videos, + audio + action trajectories

Table 2. The jump from “100M clips” to “20T tokens” reflects Cosmos 3 also ingesting text, audio and robot/human action data — not just video.

Model sizes & key hyperparameters

“Parameters” (B = billion) are the model's adjustable knobs — more usually means smarter but slower/heavier. Cosmos offers small *and* large versions so you can trade quality for speed.

Every model variant at a glance

Generation	Variants (parameters)	Helper models
Cosmos 1	Diffusion 7B & 14B; Autoregressive 4B & 12B (5B/13B for Video2World)	Tokenizer 77–105M; Prompt-up sampler 12B; Diffusion-decoder 7B
Cosmos 2	0.6B (Text2Image), 2B, 14B	Sparse-attention inference path
Cosmos 2.5	2B & 14B (each: pre-trained + post-trained); auto 7-cam & robot 3-cam variants	Cosmos-Reason1 as text encoder
Cosmos 3	Super (max accuracy), Nano (sub-second), Edge (local, soon)	Built-in reasoning transformer

Cosmos 1 diffusion hyperparameters (the most fully documented)

Setting	7B model	14B model
Transformer layers	28	36
Model (hidden) dimension	4,096	5,120
Attention heads	32	40
AdaLN-LoRA rank	256	256
Learning rate	2×10^{-4}	2×10^{-4}
Optimizer	AdamW ($\beta_1=0.9$, $\beta_2=0.99$), weight decay 0.1–0.2	
Warmup	2,500 iterations, linear	
Text encoder	T5-XXL, 512 tokens via cross-attention	
Resolution schedule	512p (57 frames) → 720p (121 frames)	
Context length @720p	56,320 tokens	
Diffusion formulation	EDM denoising score-matching; QK-RMSNorm for stability	
Compute	~10,000 H100 GPUs, ~3 months	

Table 3. NVIDIA published far more hyperparameters for Cosmos 1 than for the closed-er later releases; 2.5 and 3 reuse the same design DNA (3D-RoPE positions, AdaLN-LoRA, a causal tokenizer) at larger scale.

Inputs and outputs: the “data contract” (DTO)

Think of a **DTO** (data-transfer object) as the exact shape of what you hand the model and what it hands back. This shape widened a lot across generations.

Generation	INPUT (what you provide)	OUTPUT (what you get back)
Cosmos 1 (diffusion)	Text → T5-XXL embedding (512 tokens) + optional conditioning frames + fps + frame-count	Continuous latent video → decoded to RGB frames
Cosmos 1 (autoregressive)	Discrete video tokens (DV 8×16×16) + T5 text embedding	Next discrete tokens → diffusion-decoder → RGB frames
Cosmos 2	Text, or image, or video + fps	RGB video at 480p / 704p, 10–16 fps
Cosmos 2.5	Text / image / video, encoded by Cosmos-Reason1	Up to 30s video; optional multi-camera (3–7 synced views)
Cosmos 3	Text + image + video + ambient audio + action trajectories	Video + action trajectory + audio, preceded by explicit reasoning

Table 4. The input distribution broadens from “text + frames” (Cosmos 1) to a full five-modality stream including audio and actions (Cosmos 3). The output likewise grows from “a short video” to “video + the physical actions a robot should take”.

Why the input distribution matters

- **Cosmos 1–2:** trained mostly on curated *internet-style* video across 9 categories — great general physics, but you fine-tune for your specific robot.
- **Cosmos 2.5:** 200M *physical-AI-focused* clips + domain post-training (driving, manipulation) shift the distribution toward real deployment.
- **Cosmos 3:** adds *action trajectories from real humans and robots* and *audio* to the training mix — so the input/output distribution now includes the robot's own behaviour, not just what a camera sees.

Key results: inference speed

Speed is the headline practical difference. Faster generation = more simulations per day = faster robot/AV development. Each generation attacked latency differently.

Generation	Speed lever	Reported result
Cosmos 1	Causal, factorized tokenizer	Up to 12× faster encode/decode vs. prior tokenizer (CogVideoX)
Cosmos 2	Sparse attention in the DiT	Up to 2.6× end-to-end inference speedup
Cosmos 2.5	Flow-based sampling (fewer steps) + smaller 2B option	Faster, more resource-efficient than 2.x diffusion
Cosmos 3	Dedicated Nano variant	High-quality video + action reasoning in a fraction of a second

Table 5. Two concrete, comparable speedups are published: Cosmos 1's tokenizer (up to 12×) and Cosmos 2's sparse attention (up to 2.6×). 2.5 and 3 emphasise efficiency via flow sampling and a sub-second Nano tier rather than a single headline multiplier.

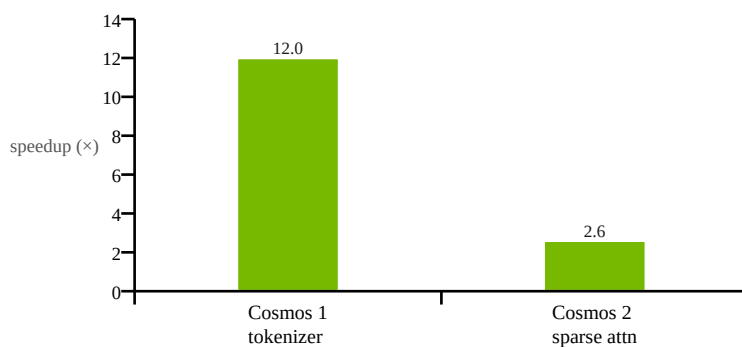


Figure 5. The two directly-quoted “× speedup” numbers. (Different things are being sped up — tokenizer vs. whole pipeline — so this compares *magnitude of improvement*, not absolute latency.)

The quality/speed trade-off in one sentence

Diffusion (Cosmos 1–2) = many denoising steps = top quality but slower; **autoregressive** (Cosmos 1) = stream tokens = fast but rougher; **flow-based** (Cosmos 2.5) = fewer steps for similar quality; **Nano** (Cosmos 3) = engineered for sub-second responses.

Pretraining and post-training: SFT and GRPO

Modern Cosmos models are built in two phases. **Pretraining** soaks up broad knowledge from massive data. **Post-training** then sharpens behaviour with curated examples (**SFT**) and reinforcement learning (**GRPO**). This recipe is clearest in **Cosmos-Reason1** — the reasoning brain that powers Cosmos 2.5's text understanding and the spirit of Cosmos 3.

The 4-stage training pipeline (Cosmos-Reason1)

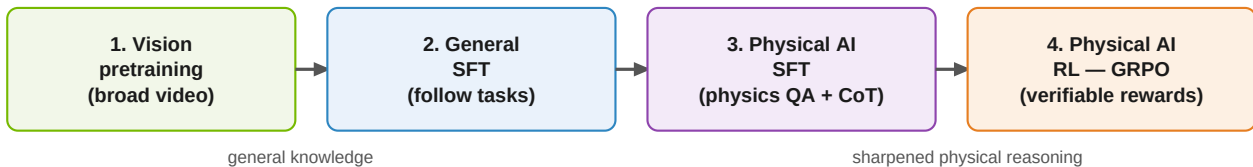


Figure 6. Knowledge first (stages 1–2), then physical-world specialisation (stage 3), then reinforcement learning to make the reasoning reliable (stage 4).

What happens in SFT (stage 3)

- Curated **question** → **answer** examples for physical common sense (space, time, basic physics) and embodied decisions (what should the robot do?).
- Built from real robotics/AV datasets: **BridgeData V2**, **RoboVQA**, **AgiBot**, **HoloAssist**, and driving data.
- The team hand-picked **~200–250 examples per data source**, turned them into multiple-choice questions, and removed ambiguous ones — quality over quantity.
- Answers include **chain-of-thought** (the model explains its reasoning step by step).

What happens in GRPO (stage 4) — in plain words

GRPO (Group Relative Policy Optimization) is reinforcement learning *without* a separate “judge” model. For each question the model produces **several attempts**; because these questions have a **verifiable correct answer**, each attempt can be auto-graded. Attempts that score **above the group average** are reinforced; below-average ones are discouraged. A **reference model** and **reward normalization** keep the updates stable so the model improves without “drifting”. The payoff: noticeably better physical reasoning.

Key experimental results

Cosmos-Reason1: bigger model & RL both help

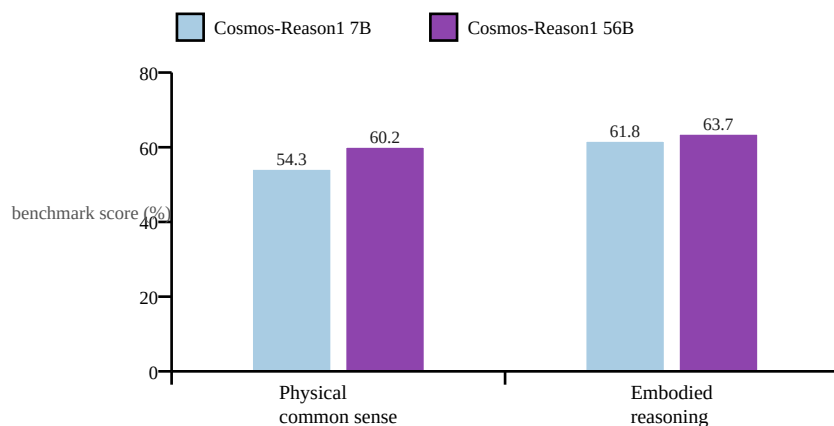


Figure 7. The 56B model beats the 7B on both physical-common-sense and embodied-reasoning benchmarks. Embodied reasoning improved by ~+10–11 points over the next-best prior model.

Reinforcement learning (GRPO) lifts reasoning further

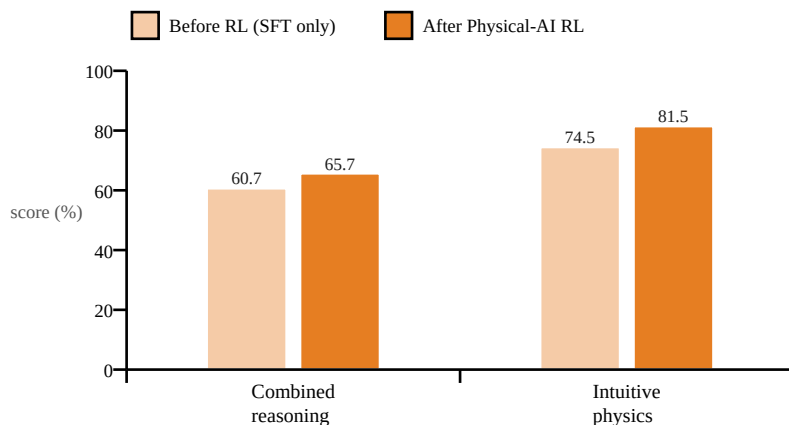


Figure 8. Adding the GRPO reinforcement-learning stage raised the 7B model by +5.0 points (combined reasoning) and +7.0 points (intuitive physics) — evidence that post-training, not just size, drives physical reasoning.

Generation-quality results (Cosmos 2.5 & 3)

- **Cosmos 2.5** post-trained **14B** scores **0.810** on PAI-Bench Image2World (matching the strong Wan2.2-A14B baseline).
- **Cosmos 2.5** autonomous-driving multiview model: up to **2.3×** better FVD/FID (video realism metrics) versus its predecessor.
- **Cosmos 3** ranks **#1 among open models** on Artificial Analysis, Physics-IQ, PAI-Bench and R-Bench (world generation); RoboLab and RoboArena (action policy); and VANTAGE-Bench and TAR (vision understanding).

Note: FVD/FID measure how realistic/consistent generated video is (lower is better, so a 2.3× improvement is large). PAI-Bench scores world-generation quality (higher is better).

Putting it all together

The four generations trace a clear arc:

- **Cosmos 1 (Toolbox):** two engines (diffusion + GPT), a strong causal tokenizer, T5 text, 9-category data, fully documented hyperparameters. Maximum flexibility.
- **Cosmos 2 (Polish):** diffusion-only, sparse attention for up to 2.6× speed, action-conditioned post-training. Same idea, executed better.
- **Cosmos 2.5 (Unify):** one flow-based model replaces three; Cosmos-Reason1 reads the prompt; 200M clips; 30s multi-camera video; RL + model-merge post-training.
- **Cosmos 3 (Leap):** mixture-of-transformers omnimodel; reasons before generating; adds audio and actions; ~20T-token training; Super/Nano/Edge tiers; #1 open model on many leaderboards.

The single sentence to remember

Cosmos went from **“many tools that dream video”** (1) → **“the best tool, polished”** (2) → **“one smart unified tool”** (2.5) → **“a reasoning, seeing, hearing, acting omnimodel”** (3).

Sources: NVIDIA *Cosmos World Foundation Model Platform* (arXiv:2501.03575); *Cosmos-Reason1: From Physical Common Sense to Embodied Reasoning* (arXiv:2503.15558); NVIDIA Research pages for Cosmos-Predict2.5 and Cosmos-Reason1; NVIDIA developer blogs for Cosmos-Predict2; Hugging Face NVIDIA blog on Cosmos Predict/Transfer 2.5; NVIDIA newsroom release for Cosmos 3 (Jun 2026). Figures are illustrative; numbers quoted from these sources. Educational summary.